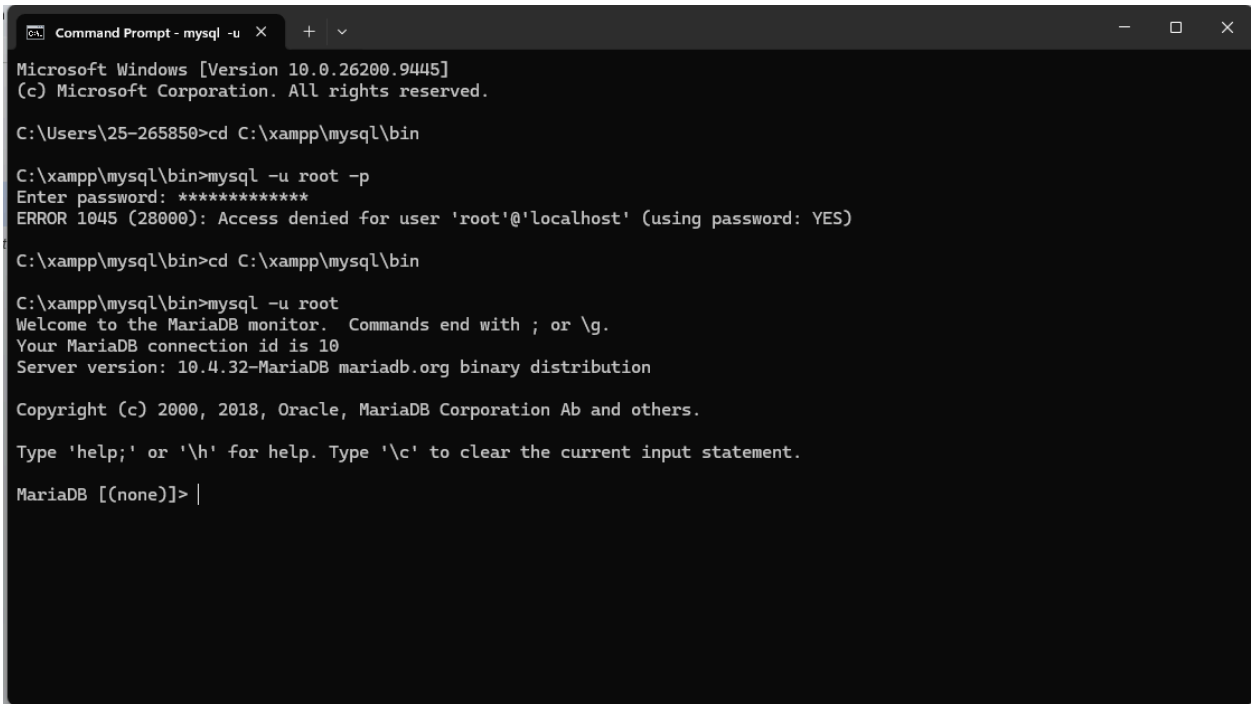
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BSIT - 304I

Document on how to access sql using cmd.

- Create database, Manipulate data for the following table:
    a. Users
    b. Students
    c. Grades
    d. Courses
- Task:
    a. insert 5 grades per student with different subject.
    b. compute the average grades per subject.
    c. compute the average grade per student
    d. show the rank of the students according to average grade



```
C:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 10
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| test               |
+--------------------+
5 rows in set (0.017 sec)

MariaDB [(none)]>
```

```
MariaDB [(none)]> CREATE DATABASE week9_DB;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| test               |
| week9_db           |
+--------------------+
6 rows in set (0.001 sec)

MariaDB [(none)]>
```

```
MariaDB [week9_DB]> CREATE TABLE Users (
    -> users_id INT AUTO_INCREMENT PRIMARY KEY,
    -> username VARCHAR(50) NOT NULL,
    -> password VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.021 sec)
```

```
MariaDB [week9_DB]> CREATE TABLE Students (
    -> students_id INT AUTO_INCREMENT PRIMARY KEY,
    -> users_id INT,
    -> s_name VARCHAR(100),
    -> s_address VARCHAR(150),
    -> FOREIGN KEY (users_id)
    -> REFERENCES Users(users_id)
    -> );
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [week9_DB]> CREATE TABLE Courses (
    -> course_id INT AUTO_INCREMENT PRIMARY KEY,
    -> c_description VARCHAR(100),
    -> c_term VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.012 sec)
```

```
MariaDB [week9_DB]> CREATE TABLE Grades (
    -> grades_id INT AUTO_INCREMENT PRIMARY KEY,
    -> students_id INT,
    -> course_id INT,
    -> grade DECIMAL(5,2),
    -> FOREIGN KEY (students_id)
    -> REFERENCES Students(students_id),
    -> FOREIGN KEY (course_id)
    -> REFERENCES Courses(course_id)
    -> );
Query OK, 0 rows affected (1.389 sec)
```

```
+-------------+---------------+---------------+
| students_id | s_name        | average_grade |
+-------------+---------------+---------------+
|           1 | James Tamaraw |     89.000000 |
|           2 | Ethan Viva    |     87.000000 |
|           3 | Karl Batoon   |     91.500000 |
|           4 | Kenneth Orton |     88.500000 |
|           5 | Warren Tatum  |     85.000000 |
+-------------+---------------+---------------+
5 rows in set (0.009 sec)
```

```
+-----------+--------------------------+---------------+
| course_id | c_description            | average_grade |
+-----------+--------------------------+---------------+
|         1 | Database Systems         |     87.600000 |
|         2 | Programming Fundamentals |     88.800000 |
+-----------+--------------------------+---------------+
2 rows in set (0.001 sec)
```

```
MariaDB [week9_DB]> SELECT
    -> s.s_name,
    -> u.username,
    -> c.c_description,
    -> c.c_term,
    -> g.grade
    -> FROM grades g
    -> INNER JOIN students s
    -> ON g.students_id = s.students_id
    -> INNER JOIN users u
    -> ON s.users_id = u.users_id
    -> INNER JOIN courses c
    -> ON g.course_id = c.course_id;
+---------------+-----------+--------------------------+----------+-------+
| s_name        | username  | c_description            | c_term   | grade |
+---------------+-----------+--------------------------+----------+-------+
| James Tamaraw | james123  | Database Systems         | 1st Term | 90.00 |
| Ethan Viva    | ethan456  | Database Systems         | 1st Term | 85.00 |
| Karl Batoon   | karl789   | Database Systems         | 1st Term | 92.00 |
| Kenneth Orton | kenneth111| Database Systems         | 1st Term | 87.00 |
| Warren Tatum  | warren222 | Database Systems         | 1st Term | 84.00 |
| James Tamaraw | james123  | Programming Fundamentals | 1st Term | 88.00 |
| Ethan Viva    | ethan456  | Programming Fundamentals | 1st Term | 89.00 |
| Karl Batoon   | karl789   | Programming Fundamentals | 1st Term | 91.00 |
| Kenneth Orton | kenneth111| Programming Fundamentals | 1st Term | 90.00 |
| Warren Tatum  | warren222 | Programming Fundamentals | 1st Term | 86.00 |
+---------------+-----------+--------------------------+----------+-------+
```